Bhuvnesh Sahu

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PROFILE

Machine Learning Engineer building production LLM, retrieval, and ML systems across enterprise RAG, tool-calling agents, calibrated classification, and edge computer vision. Statistics-trained and evaluation-first: applying leakage-checked forward-time holdouts at work, and building QLoRA/DPO/GRPO post-training pipelines for tool calling, statistical drift monitoring, and counterfactual ranking evaluation in open source.

EXPERIENCE

Wipro Bengaluru, India
ML Engineer (Account: Meta) *Jul 2025 – Present*

- Shipped incremental ingestion and hybrid retrieval for an enterprise RAG assistant on **Llama Stack** (1,200+ daily queries): structure-preserving parsers for docs, sheets, and slides with SHA-256 change detection reduced re-embedding volume by **75%**, and parent–child chunking with BM25 + dense retrieval and cross-encoder re-ranking raised top-5 recall from 61.4% (dense-only) to **84.7%** (38% relative).
- Co-designed an LLM release-readiness agent (**Llama 4 Scout**) auditing 13 go-live controls (SOX, privacy, QA, UAT) across 9 product portfolios; paired deterministic risk rules with **Pydantic**-constrained evidence checks over open tasks, unmerged code, and missing sign-offs to auto-publish weekly briefs, expanding review coverage from 7% (manual sampling) to **100%** of releases and reducing reporting time by **60%**.
- Trained and deployed a two-stage router for ∼1,100 support tickets/day (Account Management gate, then per-branch residual DNNs over dense embeddings across 433 queues). Class-weighted focal loss lifted top-1 accuracy from a 40% legacy baseline to **88%** on a forward-time holdout with post-triage leakage features removed, while a temperature-scaled confidence gate auto-routes **68%** of volume at **95%** accuracy with human fallback.
- Engineered a hierarchical 12-month OpEx forecasting pipeline (**XGBoost**, seasonal decomposition) across 80+ cost centers, achieving **4.2% MAPE** on walk-forward quarterly holdouts (26% lower error than run-rate baselines) with automated daily variance attribution for Finance.

Coriolis Technologies Pune, India
Project Intern – Applied AI (Multi-Agent Systems) *Jan 2025 – Apr 2025*

- Developed a **LangGraph** multi-agent system that plans and executes multi-step workflows across enterprise REST APIs, retrieving top-K endpoint schemas and auth rules at plan time from an embedding index over raw **OpenAPI** specifications instead of loading full specs or hand-writing wrappers.
- Orchestrated sub-task routing and **tool calling** via a supervisor agent with strict **Pydantic** schema validation; a 3-attempt self-healing retry loop that feeds validation and execution errors back to the model lifted workflow success from a 64% single-shot baseline to **92%** on a frozen 500-request benchmark.
- Reduced end-to-end workflow latency from **14 s to 4 s** with a DAG-based **asyncio** dispatcher that runs independent tool and API calls concurrently; planning ran on a locally hosted **Llama 3.1 8B** model served via **vLLM**.

Algolabs Chennai, India
Machine Learning Intern (Computer Vision & Edge AI) *May 2024 – Jul 2024*

- Built a crack instance-segmentation pipeline for 3D-printed building surfaces: curated polygon masks across **430** high-resolution inspection images in LabelMe (four severity classes: hairline, thin, wide, deep), using patch tiling to preserve fine detail and point-prompted **SAM** proposals with human review to cut annotation time by ∼**70%**.
- Benchmarked **YOLOv8-seg** against **Detectron2** (Mask R-CNN) on Tesla T4: Mask R-CNN traced fine branching cracks more cleanly but ran at 7 FPS, so selected YOLOv8-seg at **111 FPS** (0.82 precision, 0.78 mask IoU) for real-time streaming.
- Deployed real-time edge inference via **ONNX** and post-training **TensorRT INT8** quantization in **DeepStream** on NVIDIA Jetson Orin Nano (**35+ FPS**, 2.8× over FP32, <0.01 mask-IoU loss), alongside a containerized batch service on **AWS ECS**.

EDUCATION

Chennai Mathematical Institute (CMI) Chennai, India
M.Sc. in Data Science ***(CGPA: 8.31/10.0)*** *2023 – 2025*

- **Relevant Coursework:** Statistical Learning Theory, Stochastic Processes, Convex Optimization, Deep Learning.

Hindu College, University of Delhi New Delhi, India
B.Sc. (Hons.) in Statistics ***(CGPA: 8.70/10.0)*** *2020 – 2023*

- **Honors:** DST INSPIRE Scholar (top 1% of Class XII board). **Coursework:** Statistical Inference, Econometrics, Linear Algebra, Real Analysis, Probability Theory.

PROJECTS

VeriCall — Execution-Verified Post-Training for Tool Calling GitHub | Hugging Face

- Post-trained Qwen2.5-7B-Instruct for REST tool calling via **QLoRA** SFT and **DPO** on 3,200 execution-mined near-miss pairs from 36 training endpoints of a 60-endpoint FastAPI sandbox. On 500 requests across 24 held-out endpoints (4 unseen APIs), execution accuracy rose from 54.2% (base) and 77.6% (SFT) to **86.4%** (95% CI [83.1%, 89.1%], McNemar $p < 0.001$ vs. SFT), beating base + constrained decoding (68.4%).
- Cut hallucinated-parameter requests by 67% vs. SFT (58 to 19 of 500) and raised BFCL multiple-function AST accuracy ($N = 200$) from 69.5% to **82.5%**; a follow-up **GRPO** run (binary execution reward, $G = 8$) lifted Qwen2.5-1.5B-Instruct from 61.6% (SFT) to 74.8%.
- Wrote a completion-sliced PyTorch DPO trainer with precomputed reference log-probabilities, reducing logit memory from 1.83 GB to 18 MB and peak VRAM to **6.8 GB** on a 16 GB T4 (default TRL ran out of memory). Served the AWQ-quantized model on **vLLM** (86.0% accuracy retained) at **640 ms p50** end-to-end and 14 req/s on one T4 with a fixed, prefix-cached tool set (85 ms p50 TTFT, 380 ms uncached).

TraceGuard (`traceguard-ai`) — LLM Drift Monitoring & Guardrails SDK GitHub | PyPI

- Published `traceguard-ai` on **PyPI** (Pytest, GitHub Actions CI) for LLM traffic drift detection: PSI, KS, and χ^2 checks on scalar/categorical signals plus permutation-tested kernel **MMD** on prompt/response embeddings, with Benjamini–Hochberg FDR control across p-value tests, measuring 3.8% false-alarm rate under H_0 ($\alpha = 0.05$) and **89.6%** power on 25% out-of-domain semantic shift.
- Implemented async FastAPI guardrail middleware (regex/Luhn/Presidio pattern-based PII redaction, TF-IDF + heuristic prompt-injection screening, token budgets, OpenTelemetry spans) adding **3.0 ms p95** per 128-token prompt, catching **76.7%** (46/60) of injections on `deepset/prompt-injections` with 1 false positive in 56 benign prompts, and **71.2%** (712/1,000) of out-of-distribution injections on `Lakera/gandalf_ignore_instructions`.

Rankwise — Two-Stage Recommender with Counterfactual Evaluation GitHub

- Architected a two-stage recommender on KuaiRand-1K (250k+ videos after filtering): a **Two-Tower** retriever (in-batch softmax with $\log Q$ correction, FAISS HNSW) feeding a multi-task **DCN-v2** ranker that raised NDCG@10 on held-out unbiased random-exposure traffic from 0.412 (LightGBM LambdaMART) to **0.448** (paired bootstrap 95% CI $[+0.024, +0.048]$).
- Evaluated off-policy estimators against unbiased ground truth, showing naive IPS overstated policy value **2.1**× (0.384 vs. 0.182) while propensity-clipped SNIPS landed within 6.4% relative error. Served the stack behind **FastAPI** (log-replayed Redis store, ONNX Runtime, isotonic calibration lowering ECE from 0.068 to 0.014) at **11.8 ms p95** (24.5 ms p99) and 320 RPS on an 8-core CPU.

SKILLS

Languages & Tools: Python (asyncio), SQL, R, Bash, Linux, Git, Pytest

LLMs & Agents: PyTorch, Hugging Face (Transformers, TRL, PEFT), bitsandbytes, QLoRA, DPO, GRPO, vLLM, LangGraph, Llama Stack, Pydantic, RAG, Tool Calling

ML, Ranking & Retrieval: Two-Tower, DCN-v2, LightGBM (LambdaMART), XGBoost, FAISS (HNSW), BM25/Hybrid Search, Cross-Encoders, Scikit-learn, Calibration (Temperature Scaling, Isotonic), Time-Series Forecasting

Statistics & Evaluation: Hypothesis Testing (KS, χ^2, McNemar), Bootstrap/Wilson CIs, Kernel MMD, PSI, Benjamini–Hochberg FDR, Counterfactual Evaluation (IPS/SNIPS), Walk-Forward Validation

Vision, Edge & Infra: YOLOv8-seg, Detectron2 (Mask R-CNN), SAM, ONNX Runtime, TensorRT (INT8), DeepStream, NVIDIA Jetson, FastAPI, Docker, AWS ECS, Redis, OpenTelemetry, Microsoft Presidio, GitHub Actions CI/CD